GitHub Repo Link : https://github.com/khadeer329/Devops21/tree/main/Scripts

1. Create a bash script to**check if a directory is available or not**.

   Commnands Executed:

   vi Directory.bash

   cat Directory.bash

   chmod 755 Directory.bash

   ./Directory.bash

   Evidence (screenshots):

```
[root@ip-172-31-2-48 Scripts]# vi Directory.bash
[root@ip-172-31-2-48 Scripts]# cat Directory.bash
#!/bin/bash

DIR="/khadeer/Scripts"

if [ -d "$DIR" ]; then
  echo "Directory exists: $DIR"
else
  echo "Directory does not exist: $DIR"
fi
[root@ip-172-31-2-48 Scripts]# ./Directory.bash
Directory exists: /khadeer/Scripts
[root@ip-172-31-2-48 Scripts]# |
```

2. Create a bash script to**create multiple files**.

   Commnands Executed:

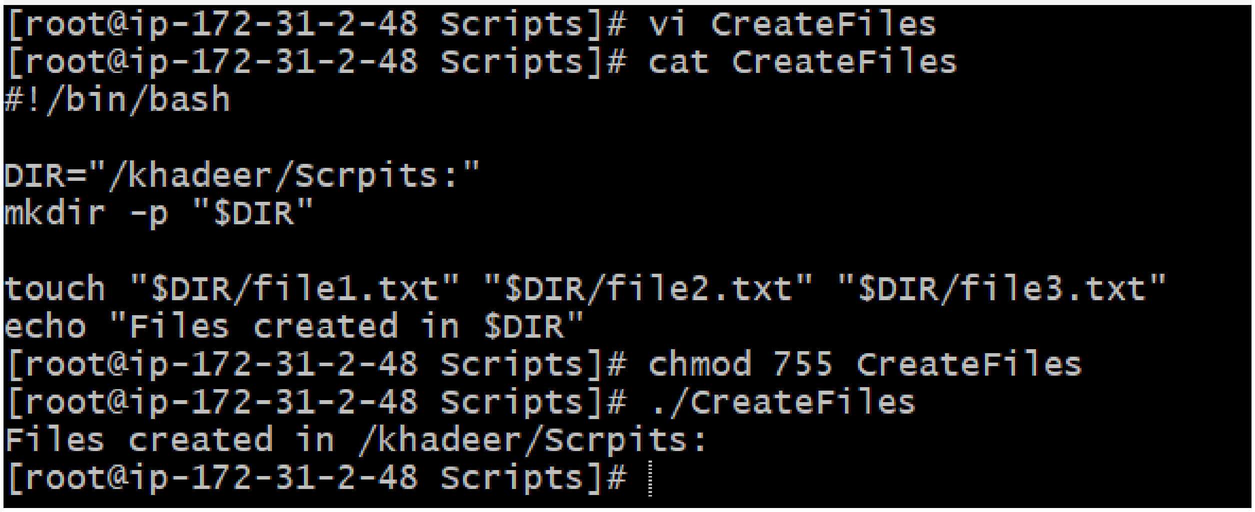
   vi CreateFiles.bash

   cat CreateFiles.bash

chmod 755 CreateFiles.bash

./CreateFiles.bash

Evidence (screenshots):



```
[root@ip-172-31-2-48 Scripts]# vi CreateFiles
[root@ip-172-31-2-48 Scripts]# cat CreateFiles
#!/bin/bash

DIR="/khadeer/Scrpits:"
mkdir -p "$DIR"

touch "$DIR/file1.txt" "$DIR/file2.txt" "$DIR/file3.txt"
echo "Files created in $DIR"
[root@ip-172-31-2-48 Scripts]# chmod 755 CreateFiles
[root@ip-172-31-2-48 Scripts]# ./CreateFiles
Files created in /khadeer/Scrpits:
[root@ip-172-31-2-48 Scripts]#
```

3. Create a bash script to**take a backup of a directory**.

Commnands Executed:

vi Backup.bash

cat Backup.bash

chmod 755 Backup.bash

./Backup.bash

Evidence (screenshots):

```
[root@ip-172-31-2-48 Scripts]# vi Backup.bash
[root@ip-172-31-2-48 Scripts]# cat Backup.bash
#!/bin/bash

SRC="khadeer/Scripts"
DEST="khadeer/Scripts"
NAME="backup_$(date +%Y%m%d_%H%M%S).tar.gz"

mkdir -p "$DEST"
tar -czf "$DEST/$NAME" "$SRC"

echo "Backup created: $DEST/$NAME"
[root@ip-172-31-2-48 Scripts]# chmod 755 Backup.bash
[root@ip-172-31-2-48 Scripts]# ./Backup.bash
tar: khadeer/Scripts: file changed as we read it
Backup created: khadeer/Scripts/backup_20260705_125201.tar.gz
[root@ip-172-31-2-48 Scripts]# ll
total 12
-rwxr-xr-x. 1 root root 186 Jul  5 12:51 Backup.bash
-rwxr-xr-x. 1 root root 140 Jul  5 12:44 CreateFiles
-rwxr-xr-x. 1 root root 140 Jul  5 12:33 Directory.bash
drwxr-xr-x. 3 root root  21 Jul  5 12:52 khadeer
[root@ip-172-31-2-48 Scripts]#
```

4. Create a bash script to**install Nginx on an EC2 server**.

Commnands Executed:

vi Eginx.bash

cat Eginx.bash

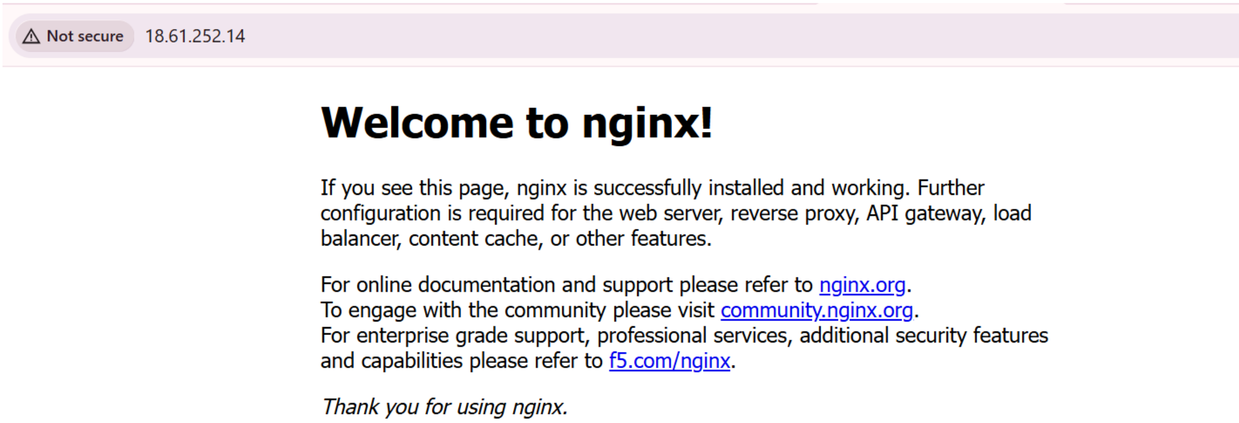
chmod 755 Eginx.bash

./Eginx.bash

Evidence (screenshots):

```
[root@ip-172-31-2-48 Scripts]# vi Nginx.bash
[root@ip-172-31-2-48 Scripts]# cat Nginx.bash
#!/bin/bash

set -e

sudo yum update -y
sudo amazon-linux-extras install nginx -y
sudo systemctl enable nginx
sudo systemctl start nginx

echo "Nginx installed and started"
[root@ip-172-31-2-48 Scripts]# chmod 755 Nginx.bash
[root@ip-172-31-2-48 Scripts]# ./Nginx.bash
Amazon Linux 2023 Kernel Livepatch repository                                                          459 kB/s |  55 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
sudo: amazon-linux-extras: command not found
[root@ip-172-31-2-48 Scripts]# vi Nginx.bash
[root@ip-172-31-2-48 Scripts]# cat Nginx.bash
#!/bin/bash

set -e

sudo yum update -y
sudo yum install nginx -y
sudo systemctl enable nginx
sudo systemctl start nginx

echo "Nginx installed and started"
[root@ip-172-31-2-48 Scripts]# ./Nginx.bash
Last metadata expiration check: 0:01:05 ago on Sun Jul  5 12:56:55 2026.
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:01:06 ago on Sun Jul  5 12:56:55 2026.
Dependencies resolved.
====================================================================================================================================
 Package                       Architecture          Version                              Repository                        Size
====================================================================================================================================
Installing:
 nginx                         x86_64                1:1.30.2-1.amzn2023.0.1              amazonlinux                       34 k
Installing dependencies:
 generic-logos-httpd           noarch                18.0.0-12.amzn2023.0.3               amazonlinux                       19 k
 gperftools-libs               x86_64                2.9.1-1.amzn2023.0.3                 amazonlinux                      308 k
 libunwind                     x86_64                1.4.0-5.amzn2023.0.3                 amazonlinux                       66 k
 nginx-core                    x86_64                1:1.30.2-1.amzn2023.0.1              amazonlinux                      709 k
```

```
Transaction Summary
====================================================================================================================================
Install  7 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Downloading Packages:
(1/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm                                           434 kB/s |  19 kB     00:00
(2/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm                                                       1.3 MB/s |  66 kB     00:00
(3/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm                                                 5.4 MB/s | 308 kB     00:00
(4/7): nginx-1.30.2-1.amzn2023.0.1.x86_64.rpm                                                          1.3 MB/s |  34 kB     00:00
(5/7): nginx-filesystem-1.30.2-1.amzn2023.0.1.noarch.rpm                                               427 kB/s |  10 kB     00:00
(6/7): nginx-core-1.30.2-1.amzn2023.0.1.x86_64.rpm                                                      18 MB/s | 709 kB     00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm                                                825 kB/s |  21 kB     00:00
------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                  8.1 MB/s | 1.1 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                            1/1
  Running scriptlet: nginx-filesystem-1:1.30.2-1.amzn2023.0.1.noarch                                                            1/7
  Installing       : nginx-filesystem-1:1.30.2-1.amzn2023.0.1.noarch                                                            1/7
  Installing       : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch                                                               2/7
  Installing       : libunwind-1.4.0-5.amzn2023.0.3.x86_64                                                                      3/7
  Installing       : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64                                                                4/7
  Installing       : nginx-core-1:1.30.2-1.amzn2023.0.1.x86_64                                                                  5/7
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                                          6/7
  Installing       : nginx-1:1.30.2-1.amzn2023.0.1.x86_64                                                                       7/7
  Running scriptlet: nginx-1:1.30.2-1.amzn2023.0.1.x86_64                                                                       7/7
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                                          1/7
  Verifying        : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64                                                                2/7
  Verifying        : libunwind-1.4.0-5.amzn2023.0.3.x86_64                                                                      3/7
  Verifying        : nginx-1:1.30.2-1.amzn2023.0.1.x86_64                                                                       4/7
  Verifying        : nginx-core-1:1.30.2-1.amzn2023.0.1.x86_64                                                                  5/7
  Verifying        : nginx-filesystem-1:1.30.2-1.amzn2023.0.1.noarch                                                            6/7
  Verifying        : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch                                                               7/7

Installed:
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch          gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64          libunwind-1.4.0-5.amzn2023.0.3.x86_64
  nginx-1:1.30.2-1.amzn2023.0.1.x86_64                       nginx-core-1:1.30.2-1.amzn2023.0.1.x86_64            nginx-filesystem-1:1.30.2-1.amzn2023.0.1.noarch
  nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Nginx installed and started
[root@ip-172-31-2-48 Scripts]# |
```



5. Create a bash script to**install Apache Tomcat on an EC2 server**.

Commnands Executed:

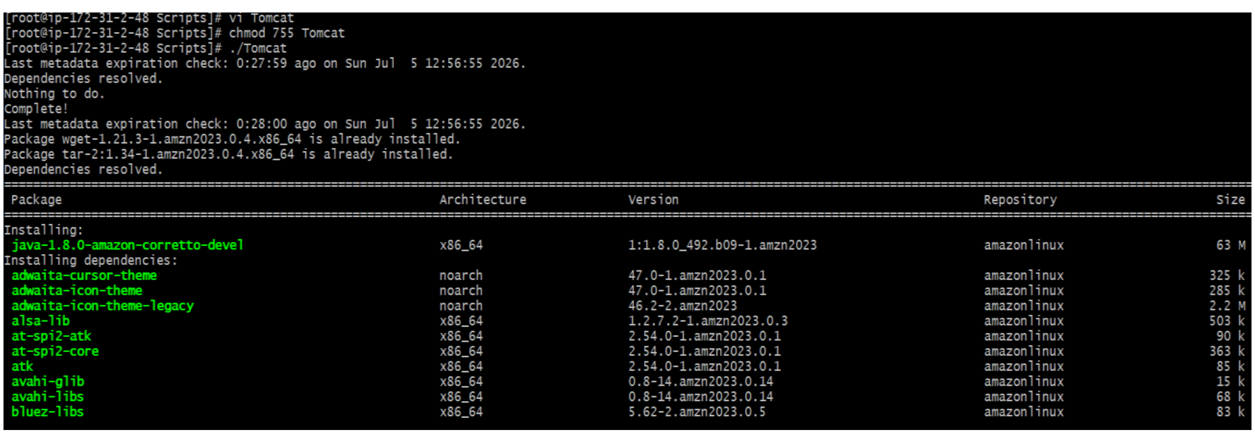
vi Tomcat

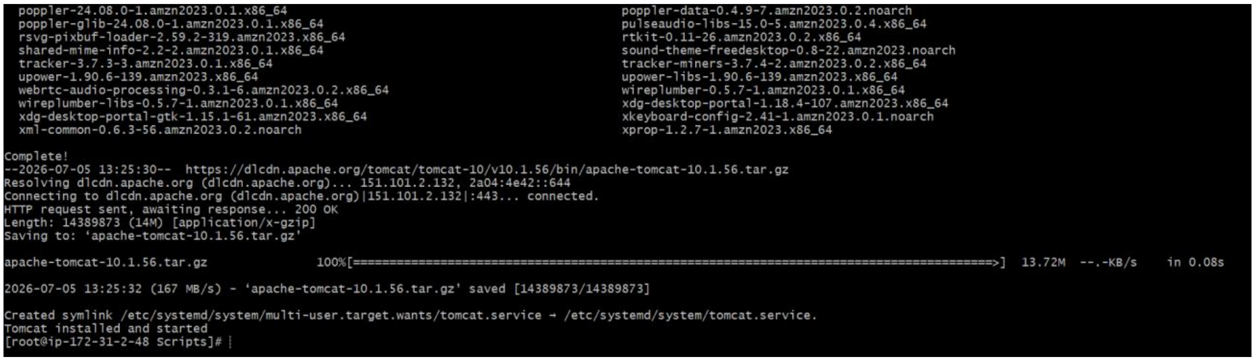
cat Tomcat

chmod 755 Tomcat

./Tomcat.bash

Evidence (screenshots):





```
  poppler-24.08.0-1.amzn2023.0.1.x86_64                              poppler-data-0.4.9-7.amzn2023.0.2.noarch
  poppler-glib-24.08.0-1.amzn2023.0.1.x86_64                         pulseaudio-libs-15.0-5.amzn2023.0.4.x86_64
  rsvg-pixbuf-loader-2.59.2-319.amzn2023.x86_64                      rtkit-0.11-26.amzn2023.0.2.x86_64
  shared-mime-info-2.2-2.amzn2023.0.1.x86_64                         sound-theme-freedesktop-0.8-22.amzn2023.noarch
  tracker-3.7.3-3.amzn2023.0.1.x86_64                                tracker-miners-3.7.4-2.amzn2023.0.2.x86_64
  upower-1.90.6-139.amzn2023.x86_64                                  upower-libs-1.90.6-139.amzn2023.x86_64
  webrtc-audio-processing-0.3.1-6.amzn2023.0.2.x86_64                wireplumber-0.5.7-1.amzn2023.0.1.x86_64
  wireplumber-libs-0.5.7-1.amzn2023.0.1.x86_64                       xdg-desktop-portal-1.18.4-107.amzn2023.x86_64
  xdg-desktop-portal-gtk-1.15.1-61.amzn2023.x86_64                   xkeyboard-config-2.41-1.amzn2023.0.1.noarch
  xml-common-0.6.3-56.amzn2023.0.2.noarch                            xprop-1.2.7-1.amzn2023.x86_64

Complete!
--2026-07-05 13:25:30--  https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.56/bin/apache-tomcat-10.1.56.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14389873 (14M) [application/x-gzip]
Saving to: ‘apache-tomcat-10.1.56.tar.gz’

apache-tomcat-10.1.56.tar.gz          100%[===================================================================================>]  13.72M  --.-KB/s    in 0.08s

2026-07-05 13:25:32 (167 MB/s) - ‘apache-tomcat-10.1.56.tar.gz’ saved [14389873/14389873]

Created symlink /etc/systemd/system/multi-user.target.wants/tomcat.service → /etc/systemd/system/tomcat.service.
Tomcat installed and started
[root@ip-172-31-2-48 Scripts]#
```

6. Create a bash script to**check if the Nginx service is running**, if not running then script should start the service.

Commnands Executed:

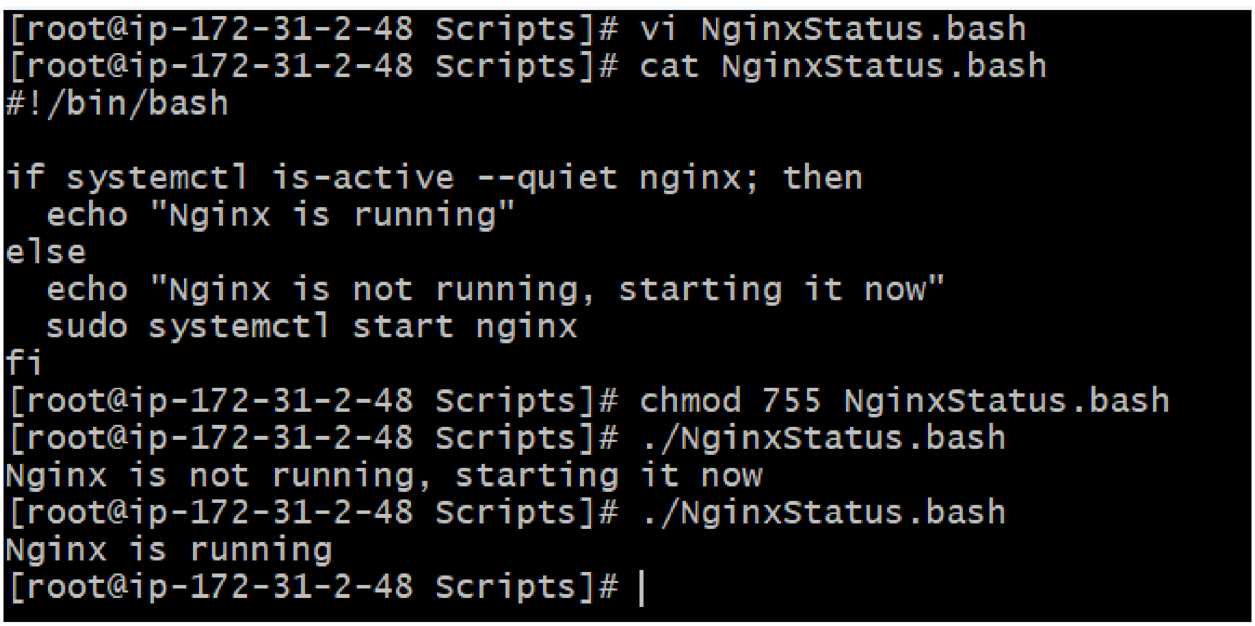
vi NginxStatus.bash

cat NginxStatus.bash

chmod 755 NginxStatus.bash

./ NginxStatus.bash

Evidence (screenshots):



```
[root@ip-172-31-2-48 Scripts]# vi NginxStatus.bash
[root@ip-172-31-2-48 Scripts]# cat NginxStatus.bash
#!/bin/bash

if systemctl is-active --quiet nginx; then
  echo "Nginx is running"
else
  echo "Nginx is not running, starting it now"
  sudo systemctl start nginx
fi
[root@ip-172-31-2-48 Scripts]# chmod 755 NginxStatus.bash
[root@ip-172-31-2-48 Scripts]# ./NginxStatus.bash
Nginx is not running, starting it now
[root@ip-172-31-2-48 Scripts]# ./NginxStatus.bash
Nginx is running
[root@ip-172-31-2-48 Scripts]# |
```

7. Create a bash script for a**calculator**.

Commnands Executed:

vi Calculator.bash

cat Calculator.bash

chmod 755 Calculator.bash

./Calculator.bash

Evidence (screenshots):

```
[root@ip-172-31-2-48 Scripts]# vi Calculator.bash
[root@ip-172-31-2-48 Scripts]# cat Calculator.bash
#!/bin/bash

read -p "Enter first number: " a
read -p "Enter operator (+ - * /): " op
read -p "Enter second number: " b

case "$op" in
  +) result=$((a + b)) ;;
  -) result=$((a - b)) ;;
  \*) result=$((a * b)) ;;
  /) result=$((a / b)) ;;
  *) echo "Invalid operator"; exit 1 ;;
esac

echo "Result: $result"
[root@ip-172-31-2-48 Scripts]# chmod 755 Calculator.bash
[root@ip-172-31-2-48 Scripts]# ./Calculator.bash
Enter first number: 5
Enter operator (+ - * /): *
Enter second number: 5
Result: 25
[root@ip-172-31-2-48 Scripts]# |
```

8. Create a bash script to**check if a directory exists**, if not then create a directory.

Commnands Executed:

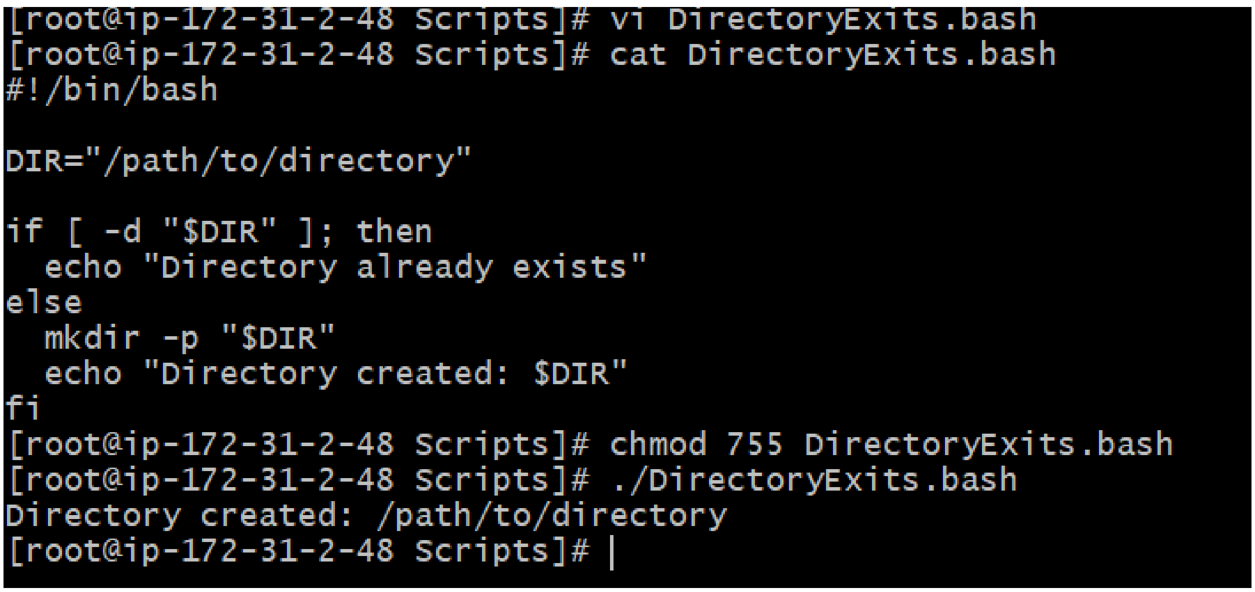
vi DirectoryExits.bash

cat DirectoryExits.bash

chmod 755 DirectoryExits.bash

./ DirectoryExits.bash

Evidence (screenshots):



```
[root@ip-172-31-2-48 Scripts]# vi DirectoryExits.bash
[root@ip-172-31-2-48 Scripts]# cat DirectoryExits.bash
#!/bin/bash

DIR="/path/to/directory"

if [ -d "$DIR" ]; then
  echo "Directory already exists"
else
  mkdir -p "$DIR"
  echo "Directory created: $DIR"
fi
[root@ip-172-31-2-48 Scripts]# chmod 755 DirectoryExits.bash
[root@ip-172-31-2-48 Scripts]# ./DirectoryExits.bash
Directory created: /path/to/directory
[root@ip-172-31-2-48 Scripts]# |
```

9. Create a bash script to**delete the last 3 lines of a file**.

Commnands Executed:

vi DeleteFile.bash

cat DeleteFile.bash

chmod 755 DeleteFile.bash

./DeleteFile.bash

Evidence (screenshots):

```
[root@ip-172-31-2-48 Scripts]# vi DeleteFile.bash
[root@ip-172-31-2-48 Scripts]# cat  DeleteFile.bash
#!/bin/bash

FILE="/path/to/file.txt"
LINES=$(wc -l < "$FILE")

if [ "$LINES" -le 3 ]; then
  > "$FILE"
else
  head -n -3 "$FILE" > "${FILE}.tmp" && mv "${FILE}.tmp" "$FILE"
fi

echo "Last 3 lines removed"
[root@ip-172-31-2-48 Scripts]# chmod 755 DeleteFile.bash
[root@ip-172-31-2-48 Scripts]# ./DeleteFile.bash
./DeleteFile.bash: line 4: /path/to/file.txt: No such file or directory
./DeleteFile.bash: line 6: [: : integer expression expected
head: cannot open '/path/to/file.txt' for reading: No such file or directory
Last 3 lines removed
[root@ip-172-31-2-48 Scripts]# |
```

10. Bash script to monitor cpu and if it is more than 80% then send email notification.

Commnands Executed:

vi CpuAlert.bash

cat CpuAlert.bash

chmod 755 CpuAlert.bash

./CpuAlert.bash

Evidence (screenshots):

```
[root@ip-172-31-2-48 Scripts]# vi CpuAlert.bash
[root@ip-172-31-2-48 Scripts]# cat CpuAlert.bash
#!/bin/bash

THRESHOLD=80
EMAIL_TO="khadeershaik0329@gmail.com"

CPU=$(top -bn1 | awk '/Cpu\(s\)/ {print 100 - $8}')

CPU_INT=${CPU%.*}

if [ "$CPU_INT" -gt "$THRESHOLD" ]; then
  echo "High CPU usage: ${CPU_INT}%" | mail -s "CPU Alert" "$EMAIL_TO"
fi
[root@ip-172-31-2-48 Scripts]# chmod 755 CpuAlert.bash
[root@ip-172-31-2-48 Scripts]# ./CpuAlert.bash
```

11. Bash script to monitor disk space and if it is more than 80% then send email notification.

Commnands Executed:

vi DiskAlert.bash

cat DiskAlert.bash

chmod 755 DiskAlert.bash

./DiskAlert.bash

Evidence (screenshots):

```
[root@ip-172-31-2-48 Scripts]# vi DiskAlert.bash
[root@ip-172-31-2-48 Scripts]# cat DiskAlert.bash
#!/bin/bash

THRESHOLD=80
EMAIL_TO="you@example.com"

DISK=$(df / | awk 'NR==2 {gsub("%","",$5); print $5}')

if [ "$DISK" -gt "$THRESHOLD" ]; then
  echo "High disk usage: ${DISK}%" | mail -s "Disk Alert" "$EMAIL_TO"
fi
[root@ip-172-31-2-48 Scripts]# chmod 755 DiskAlert.bash
[root@ip-172-31-2-48 Scripts]# ./Di
Directory.bash       DirectoryExits.bash  DiskAlert.bash
[root@ip-172-31-2-48 Scripts]# ./DiskAlert.bash
```

12 . Bash script to monitor memory and if it is more than 80% then send email notification.

Commnands Executed:

vi MemoryAlert.bash

cat MemoryAlert.bash

chmod 755 MemoryAlert.bash

./MemoryAlert.bash

Evidence (screenshots):

```
[root@ip-172-31-2-48 Scripts]# vi MemoryAlert.bash
[root@ip-172-31-2-48 Scripts]# cat MemoryAlert.bash
#!/bin/bash

THRESHOLD=80
EMAIL_TO="you@example.com"

MEM=$(free | awk '/Mem:/ {printf("%.0f", $3/$2 * 100)}')

if [ "$MEM" -gt "$THRESHOLD" ]; then
  echo "High memory usage: ${MEM}%" | mail -s "Memory Alert" "$EMAIL_TO"
fi
[root@ip-172-31-2-48 Scripts]# chmod 755 MemoryAlert.bash
[root@ip-172-31-2-48 Scripts]# ./MemoryAlert.bash
[root@ip-172-31-2-48 Scripts]#
```

13.Crontab

Create a file named crontab.txt :

vi crontab.txt

```
#purpose Crontab
#owner : Srihari

# 1. April 5th at midnight
0 0 5 4 * echo "April 5th midnight"

# 2. 5th of Jan, Jun, Nov - only if Thursday
0 0 5 1,6,11 4 echo "5th of Jan/Jun/Nov and Thursday"

# 3. At minutes 05 and 27 of hours 9, 10, 11
5,27 9,10,11 * * * echo "Minutes 05 and 27 of 9-11"

# 4. 34th minute of 9AM on August 15th
34 9 15 8 * echo "August 15th 9:34 AM"

# 5. Every midnight
0 0 * * * echo "Midnight every day"

# 6. Every Saturday at 11:59 PM
59 23 * * 6 echo "Saturday night 11:59 PM"

# 7. After every reboot
@reboot echo "System rebooted"
```

Install Cron:

sudo yum install cronie -y

```
[root@ip-172-31-18-95 ec2-user]# sudo yum install cronie -y
Last metadata expiration check: 6:11:53 ago on Sun Jul  5 09:05:55 2026.
Dependencies resolved.
================================================================================================
 Package               Architecture     Version                        Repository          Size
================================================================================================
Installing:
 cronie                x86_64           1.5.7-1.amzn2023.0.2           amazonlinux        115 k
Installing dependencies:
 cronie-anacron        x86_64           1.5.7-1.amzn2023.0.2           amazonlinux         32 k
```

Step 3: Start the Cron Service

sudo systemctl start crond

Enable it at boot:

sudo systemctl enable crond

Check its status:

sudo systemctl status crond

output:

Active: active (running)

```
[root@ip-172-31-18-95 ec2-user]# sudo systemctl start crond
[root@ip-172-31-18-95 ec2-user]# sudo systemctl enable crond
[root@ip-172-31-18-95 ec2-user]# sudo systemctl status crond
● crond.service - Command Scheduler
     Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-07-05 15:18:28 UTC; 16s ago
```

Step 4: Verify crontab is Available:

crontab -l

If you haven't created one yet, you'll see:

no crontab for root

This is normal.

Step 5: Install Your Cron File

crontab crontab.txt

Verify:

crontab -l

```
[root@ip-172-31-18-95 ec2-user]# crontab crontab.txt
[root@ip-172-31-18-95 ec2-user]# crontab -l
#purpose Crontab
#owner : Srihari

# 1. April 5th at midnight
0 0 5 4 * echo "April 5th midnight"

# 2. 5th of Jan, Jun, Nov - only if Thursday
0 0 5 1,6,11 4 echo "5th of Jan/Jun/Nov and Thursday"

# 3. At minutes 05 and 27 of hours 9, 10, 11
5,27 9,10,11 * * * echo "Minutes 05 and 27 of 9-11"

# 4. 34th minute of 9AM on August 15th
34 9 15 8 * echo "August 15th 9:34 AM"

# 5. Every midnight
0 0 * * * echo "Midnight every day"

# 6. Every Saturday at 11:59 PM
59 23 * * 6 echo "Saturday night 11:59 PM"

# 7. After every reboot
@reboot echo "System rebooted"
```